

SOUMYA K

M.Tech (Power Electronics)

Experienced Power Electronics Design Engineer| Bangalore, Karnataka |

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PROFESSIONAL SUMMARY:

Power Electronics design engineer with 3 years of experience in designing DC-DC Power supplies for Space, Navy and Automotive application. Proficient in designing DC-DC Converter, SMATH Design calculations, ORCAD Schematics Design. Seeking for the position of Electronics design engineer in research and development area where my skills, experience and knowledge of updated electronics design techniques would boost the development of electronics products in the organization

TECHNICAL SKILLS:

- Switched mode power supply design.
- Topology worked on Forward Converters.
- Familiar with OrCAD software in drawing schematics.
- Experience in circuit simulation tools like LT Spice.
- Familiar with SMATH and MATHCAD software for doing design calculations.
- Hands on Experience with DC/DC converters
- Tools: OrCAD, SMATH, AP300 Equipment with Frequency Analyzer, LTspices.
- Both practical and theoretical knowledge in various electronic devices.
- Identifying cause of failure in electronic components and providing preventive solution on the same.
- Circuit debugging, Fault analyzing.
- Experience in carrying out Derating, FMEA (Failure Mode Effects Analysis) etc.
- Hand's on DSO, AP300 (frequency response analyzer) instruments, Programmable supplies, Electronic Load, DMMs etc.

EDUCATION QUALIFICATIONS:

DR. Ambedkar Institute of Technology, Bengaluru, Karnataka | Masters Technology (M.TECH) In Power Electronics. PERCENTAGE: 89.1%

Brindavan College of Engineering, Bengaluru, Karnataka | Bachelor of Engineering (BE). In Electronics and Communication Engineering PERCENTAGE: 65.77%

INDUSTRIAL EXPERIENCE:

- **CENTUM ELECTRONICS LTD, BENGALURE**

Intern-Power Electronics Design | July-2021 – July-2022|

- **CENTUM ELECTRONICS LTD, BENGALURE**

Engineer-Power Electronics Design | Aug-2022 – Present |

RESPONSIBILITIES:

- Design of DC-DC converters which includes Topology Selection, MOSFET'S, Magnetics, Capacitors, PWM IC, Switching Frequency, Driver Circuit, Feedback method and required Protection circuits.
- Good experience in design and bench level testing in MOSFET, IGBT, Transistor, Diode, Op-Amp's.
- Hands on experience in using Oscilloscope, frequency generator, Multimeter, Power analyser & other EE lab tools etc.
- Familiar with OrCAD software in drawing schematics.
- Familiar with SMath and MATHCAD software for doing engineering calculations.
- Experience in circuit simulation tools LTspice, Pspice.
- Experience in carrying out Derating, FMEA(Failure Mode Effects Analysis) etc.
- Coordinate with mechanical team, mass and height analysis, power dissipation of each component for finalization of mechanical package.
- Preparing critical and preliminary design review documents as per standards.
- Preparation of BOM, MATHCAD files, test plans and test reports.
- Preparing RFQ'S and Realization Report for DC-DC converters.
- Coordinating with PCB team for layout guidance and mechanical team for thermal analysis.
- Assembling the Components on the PCB (Soldering).
- Working on improving the performance of converter like load regulation, cross regulation, line regulation, loop stability, efficiency and ripple.
- Using AP300 instrument, tuning phase and gain margin to achieve the desired stability level.
- Knowledge on EMI/EMC test standard MIL-PRF- 461E.

MAJOR PROJECTS:

- **70V bus 40W triple output MOSFET based forward converter for space application (R&D project)**

Role: Design Engineer, Design & Development.

Design Tool: OrCAD, MATHCAD, LTSpice

Customer: URSC SPACE

Description: The work is focused on design and development of 70V bus 40W triple output converter which operates over the input voltage range of 60-100V DC. The output's Voltage of the converter is 5V/4, 15V/0.66, 15V/0.66A VDC.

- **40V Bus 48W triple output MOSFET based forward converter for space application (R&D project)**

Role: Design Engineer, Design & Development.

Design Tool: OrCAD, MATHCAD, LTSpice

Customer: URSC SPACE

Description: The work is focused on design and development of 40V bus 40W triple output converter which operates over the input voltage range of 18-50V DC. The output's Voltage of the converter is 5V/4, 15V/0.66, 15V/0.66A VDC. Developed detailed test plans and procedures specific to the multi-output forward converter including tests for voltage regulation, load response, protection test (OCP, OVP & UVP) and efficiency. Performed comprehensive testing to validate the forward converter's performance, including: Voltage Regulation, Cross-Regulation, Efficiency Testing.

- **Power supply requirements for RF and Digital systems (RFQ)****Description:** Selected micro buck modules with respect to the power supply requirements of RF and digital systems. Selected capacitors and inductor for optimal performance

- **Multi-output Mag-amp controlled forward converter for space application (Product order).**

Role: Design Engineer, Design & Development.

Design Tool: OrCAD, MATHCAD, LTSpice

Customer: URSC SPACE

Description: Designed circuit using OrCAD software.

- Prepared design calculation which includes inductor and capacitor to select output filter, area product calculation to select transformer and power dissipation calculation of power components like rectifier diode, MOSFET and inductor power loss using S-MATH.
- Component selection like MOSFET, Rectifier diode, DC-link capacitor, Gate drivers, feedback generator IC and PWM IC.
- Designed post regulator Mag-amp circuit and discrete low drop out LDO's (both MOSFET and Transistor based) to obtain desired outputs for optimal performance.
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LANGUAGES KNOWN:

English(Fluent), Kannada (Native), Telugu(Fluent) , Hindi (Fluent)

